Critical Site Percolation: A Conditional Theorem

What is proved, what is assumed, and what is machine-checked

3 September 2026

Abstract

This note is about Bernoulli site percolation at its critical parameter. It states two implications, both proved and machine-checked in Lean, and it states precisely what a proof that $\theta(p_c) = 0$ still rests on. First, site percolation is exactly a model of independent hyperedges, so the finite inequality for hyperedges yields the corresponding inequality for sites with no vertex forced open. Second, site percolation is monotone in the vertex set of an induced subgraph, from which the half-space theorem gives that no slab percolates at the critical parameter, and the critical statement then follows from the same-parameter slab reduction. Two ingredients remain: the finite inequality, which is the new statement, and the slab reduction. One theorem is imported, the site form of the half-space theorem.

1 The model and the target

Let G be a graph. In site percolation each vertex is open independently, with probability p_v at v, and $x \leftrightarrow y$ means that some path from x to y has all of its vertices open. A closed vertex is therefore connected to nothing, not even to itself. Write $C(x)$ for the set of vertices connected to x.

On $\mathbb{Z}^d$ with the common parameter p, let

$$\theta(p) = \mathbb{P}_p(C(0) \text{ is infinite}), \qquad p_c = \inf\{p : \theta(p) > 0\}\,.$$

Definition 1.1. The target is the assertion that $\theta(p_c) = 0$ for every $d \geq 2$.

2 The finite input

A model of independent hyperedges consists of a finite vertex set, a finite set of labels, an incidence set of vertices for each label, and an opening probability for each label, the labels being open independently. Two vertices are connected when a chain of open labels joins them; here connection is reflexive, so every vertex is connected to itself.

Definition 2.1. The finite input is the following inequality. Let a model of independent hyperedges be given, let A be a nonempty finite set of vertices, and let o and b be vertices. Then

$$\mathbb{P}(o \leftrightarrow b,\ o \leftrightarrow A) \ \geq\ \mathbb{P}(o \leftrightarrow A) \min_{a \in A} \mathbb{P}(a \leftrightarrow b)\,. \tag{1}$$

For incidence sets of size two the model is bond percolation, and dropping the event $\{o \leftrightarrow A\}$ from the left of (1) gives the inequality that Kozma and Nitzan conjectured, which is proved and machine-checked for that case.

3 Site percolation is a model of hyperedges

Given a graph and site probabilities, take one hypergraph vertex for each site and one for each edge, the latter called a port. Take one label for each site, open exactly when that site is open, and let the label of v be incident to v and to the port of every edge at v.

Since the labels are the sites, the two models have the same configuration space and the same measure. The content of the representation is that they also have the same connection events.

Lemma 3.1. *Let x and y be distinct sites. Then the event that a chain of open labels joins x to y is the event that a path of open sites joins x to y.*

Proof. Adjacency in the open hypergraph never joins two sites directly, because a label incident to two distinct sites would have to be each of them. It joins a site to a port at that site, or two ports at a common site, and in both cases the label witnessing it is the label of that site, which is therefore open.

Suppose a path of open sites joins x to y. Each step from a to b uses an edge of the graph with both endpoints open, so the labels of a and of b are open and both are incident to the port of that edge. Hence a and b are joined through that port, and the steps compose.

Conversely, suppose a chain of open labels joins x to y. Carry along the chain the assertion that a site reached is joined to x by a path of open sites, and that a port reached has an open endpoint which is so joined. The three cases of the previous paragraph preserve that assertion, and it holds at the start. At y it says that a path of open sites joins x to y. Since x and y are distinct, the first step of that path shows that x is open. $\square$

Theorem 3.2. *Assume the finite input of Definition 2.1. Let a finite graph with site probabilities be given, let A be a nonempty set of vertices, and let o and b be distinct vertices, neither of them in A. Then*

$$\mathbb{P}(o \leftrightarrow b,\ o \leftrightarrow A) \ \geq\ \mathbb{P}(o \leftrightarrow A) \min_{a \in A} \mathbb{P}(a \leftrightarrow b)\,.$$

Proof. Apply the finite input to the hypergraph above, with the set A and the vertices o and b. The hypotheses make each of the pairs (o, b), (o, a) and (a, b) distinct for $a \in A$, so Lemma 3.1 identifies every event in the inequality with its counterpart for sites, and the two measures coincide. $\square$

No vertex has to be forced open. There is also a form in which the terminals are pinned; that hypothesis is needed only to include coincident terminals, because connection is reflexive in the hypergraph while a closed site is connected to nothing.

4 From the reduction to the critical statement

The lattice half of the argument is a reduction: percolation on $\mathbb{Z}^d$ at p forces percolation, at the same p, in a slab of some finite width. One way to close the argument is to translate the slab into a half-space and quote a half-space theorem. The following observation replaces that step by a weaker one.

Definition 4.1. The reduction is the assertion that if $\theta(p) > 0$, then site percolation at the same parameter p percolates in a slab of some finite width. The slab assumption is the assertion that no slab percolates at p_{c}.

Theorem 4.2. *Assume the reduction and the slab assumption. Then $\theta(p_{\mathrm{c}}) = 0$.*

Proof. The percolation probability is nonnegative, being the measure of an event. If $\theta(p_c)$ were nonzero, then it would be positive, and the reduction would give a slab that percolates at p_c, which the slab assumption forbids. □

Theorem 4.3 (monotonicity in the underlying graph)**.** *Let G be a graph and let f be an injection of a countable vertex set into the vertices of G carrying adjacency to adjacency. Then site percolation on the pullback graph is dominated by site percolation on G: for every vertex x of the smaller graph and every p,*

$$\theta_{f^*G}(x,p) \;\leq\; \theta_G(f(x),p).$$

In particular, for $S \subseteq T$ and $x \in S$, $\theta_{G[S]}(x,p) \leq \theta_{G[T]}(x,p)$.

Proof. Restricting a configuration along f carries the site measure on the vertices of G to the site measure on the smaller vertex set, because the coordinates indexed by the image are independent with the same parameter and f is injective. If two vertices are joined by a path all of whose vertices are open in the restricted configuration, their images are joined by a path all of whose vertices are open, so f maps the open cluster of x injectively into the open cluster of $f(x)$, and an infinite cluster maps to an infinite cluster. The event that the open cluster of a vertex is infinite is measurable: a connection is a countable union over the lists of vertices witnessing an open walk, each of which is a finite intersection of one-vertex events, and the cluster is infinite exactly when it leaves every finite set of vertices. □

Corollary 4.4. *The half-space theorem implies the slab assumption: if site percolation in $\{x : x_1 \geq 0\}$ has no infinite cluster at p_c, then no slab has one either.*

Proof. A slab $\{x : 0 \leq x_1 \leq k\}$ is contained in $\{x : x_1 \geq 0\}$, and both graphs are induced from $\mathbb{Z}^d$, so Theorem 4.3 applies. As θ is nonnegative, the slab value is zero. □

5 What remains

The target of Definition 1.1 follows by Theorems 3.2, 4.2 and 4.3 from three statements:

1. the finite input of Definition 2.1;
2. the reduction of Definition 4.1;
3. the half-space theorem for site percolation.

The first two are open. The first is the new finite statement, and it is the one that carries the novelty: it is the hyperedge form of what Kozma and Nitzan conjectured. The second is the geometric heart of the lattice argument, and its difficulty is identified in the literature. Grimmett and Marstrand prove that if $\theta(p) > 0$ and $\gamma > 0$ then $p + \gamma > p_c(S(k))$ for some k, and write that they cannot remove the γ, the difficulty lying in their use of sprinkling, that is, of adding a small density of extra open sites to create open paths. The reduction of Definition 4.1 is exactly the statement with γ removed, and replacing sprinkling by the finite input of Definition 2.1 is the point of the reduction stated here.

The third is a theorem, in two published pieces, each covering site percolation. Barsky, Grimmett and Newman prove continuity of the percolation probability for independent nearest-neighbour bond or site percolation on a half-space, at the critical density *of the half-space*. Grimmett and Marstrand prove, for site percolation on $\mathbb{Z}^d$ with $d \geq 3$, that $p_c = p_c(S)$ and hence $p_c = p_c(\mathbb{H})$. Neither alone gives the statement used here, which places the half-space percolation probability at the critical parameter of the full lattice.

Remark 5.1. The half-space theorem can be dispensed with. Suppose the reduction is strengthened so that $\theta(p) > 0$ for $0 < p < 1$ gives slab percolation at some parameter $q < p$. A slab lies in $\mathbb{Z}^d$, so Theorem 4.3 gives $\theta(q) > 0$; if $p = p_c$ this contradicts the definition of p_c as an infimum. The conclusion then rests on no theorem about percolation at a critical point, and on neither published piece above. It rests instead on a reduction strictly stronger than the one in Definition 4.1.

6 The Lean formalization

Everything above is formalized. The site model, the connection events, the percolation probability and the critical parameter are the declarations `siteConn`, `thetaSite` and `criticalProbSiteI`. The critical parameter is proved to lie in the unit interval, so the target is the plain equation `thetaSite` d (`criticalProbSiteI` d) $= 0$ and carries no hypothesis under which it could hold vacuously. Six further lemmas pin the conventions: a closed vertex is connected to nothing, not even to itself; the cluster of a closed vertex is empty; every vertex of a cluster is open; and two adjacent open vertices are connected. The three remaining statements are `HyperedgeGluing`, `SiteSlabReduction` and `SiteHalfSpaceCriticality`. The last is the conjunction of

`SiteHalfSpaceOwnCriticality`, `SiteHalfSpaceCriticalEqual`,

through `siteHalfSpaceCriticality_of_published`. The strengthened reduction and the endgame that consumes it are

`SiteSlabReductionBelow`, `siteCriticality_of_slabReductionBelow`.

The proved implications are

`hyperConn_eq_siteConn`, `siteGluingUnpinned_of_hyperedgeGluing`,

`thetaSiteOn_comap_le`, `thetaSiteOn_induce_mono`,
`siteSlabCriticality_of_halfSpace`,

`siteCriticality_of_slabCriticality`, `siteCriticality_of_halfSpace`.

The measurability of the infinite-cluster event is `measurableSet_siteInfinite`, and the fact that restriction carries the site measure to the site measure is `siteBernoulli_map_restrictSite`. Each compiles under Lean 4.32.0 with the Mathlib revision of the accompanying development, with no `sorry`, and each depends on exactly the axioms `propext`, `Classical.choice` and `Quot.sound`.

The three remaining statements appear as named hypotheses, so any proof that uses them displays them. A formalization that introduces the half-space statement as an axiom proves a conditional statement, and should be described as such.

References

- D. J. Barsky, G. R. Grimmett and C. M. Newman, Percolation in half-spaces: equality of critical densities and continuity of the percolation probability, *Probab. Theory Related Fields* **90** (1991), 111–148.
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